HRV Interpretation Across Physiological Tasks Using Single-Lead ECG

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***Abstract*—Heart rate variability (HRV) is widely used to assess autonomic regulation, but standard metrics can change meaning across physiological tasks. This single-participant lab study analyzed 14 single-lead ECG recordings (500 Hz) across four experimental conditions: postural baseline (supine/sitting/standing), voluntary breath-hold (inspiratory and expiratory, two trials each), walking with recovery, and paced breathing at five rates (12, 9, 6, 5, 3 breaths/min). Postural loading produced monotonic vagal withdrawal: mean HR rose from 54.0 to 79.7 bpm and RMSSD fell from 99.1 to 25.2 ms. Inspiratory breath-hold produced larger HR excursions than expiratory hold (ΔHR up to +14.3 vs. +1.9 bpm). Walking elevated HR to 83.0 bpm with rapid post-exercise recovery ($\tau = 6.8$ s), but motion artifacts limited fine-scale HRV interpretation. Paced breathing entrained the RR spectral peak, which closely matched the imposed breathing frequency ($R^2 = 0.998$); LF/HF inflated from 0.22 to 17.47 while RMSSD remained stable (47.7–64.9 ms), indicating spectral-band migration rather than sympathovagal reversal. We report a condition-by-condition summary of which HRV metrics remained interpretable in this dataset. These results indicate that HRV metrics should be interpreted with reference to task physiology and signal-quality constraints.**

***Index Terms*—Heart rate variability, single-lead ECG, paced breathing, breath-hold, spectral band migration, autonomic nervous system**

I. Introduction

Heart rate variability quantifies beat-to-beat fluctuations in cardiac cycle length, driven primarily by autonomic neural modulation of the sinoatrial node [**?**]. Time-domain indices such as SDNN and RMSSD capture overall and short-term variability, while frequency-domain decomposition into low-frequency (LF: 0.04–0.15 Hz) and high-frequency (HF: 0.15–0.40 Hz) bands provides a spectral view of autonomic oscillations [**?**]. The HF band primarily reflects respiratory sinus arrhythmia (RSA), the phasic vagal modulation of heart rate coupled to the breathing cycle [**?**]. HRV has become a standard noninvasive tool in cardiovascular research and clinical risk stratification [**?**], [**?**].

Although HRV is widely used, the physiological interpretation of a given metric depends on the recording condition. The LF/HF ratio, originally proposed as a sympathovagal balance index [**?**], has been challenged on mechanistic grounds: LF power reflects both sympathetic and parasympathetic modulation, and the ratio becomes particularly unreliable when respiratory frequency falls below the conventional 0.15 Hz boundary separating LF from HF [**?**], [**?**]. Under such conditions, the dominant respiratory oscillation migrates into the LF band, inflating LF/HF without an actual shift in autonomic balance. This limitation is clinically relevant because slow-breathing interventions are increasingly used in biofeedback therapy [**?**].

Here, we examine how common HRV metrics behave across four task contexts: steady-state posture, transient breath-hold, ambulatory motion, and externally paced respiration. The study was designed as a within-subject lab investigation to identify conditions in which standard HRV indices remain interpretable or become misleading. We acquired 14 single-lead ECG recordings (500 Hz) from one participant using a unified analysis pipeline spanning (1) postural baseline, (2) voluntary breath-hold, (3) ambulatory walking with recovery, (4) paced breathing at five rates. This design removes between-subject variability from the comparison and allows within-participant assessment of how the *same* HRV indices behave under different physiological demands.

We formulated five hypotheses prior to data collection:

- **H1**: Progressive orthostatic loading (supine $\rightarrow$ sitting $\rightarrow$ standing) produces monotonic increases in HR and decreases in vagal-linked indices (RMSSD, HF power).
- **H2**: Inspiratory breath-hold imposes a greater chronotropic challenge than expiratory breath-hold, consistent with altered thoracic mechanics and autonomic compensation.
- **H3**: Walking-phase fine-scale HRV (RMSSD) is contaminated by motion artifacts, whereas mean HR and post-exercise recovery kinetics remain interpretable.
- **H4**: Within the tested paced-breathing range (3–12 breaths/min), RR oscillation amplitude does not exhibit a classical resonance maximum centered at 6 breaths/min.
- **H5**: When breathing frequency falls below 0.15 Hz, the resulting LF/HF inflation reflects spectral-band migration, not sympathovagal shift, and therefore dissociates from RMSSD.

Note that successful entrainment of the RR spectral peak by paced breathing (verified by linear regression of measured vs. imposed frequency) is treated as a manipulation check rather than a hypothesis, since respiratory entrainment of the cardiac cycle is well-established.

The analysis has three components. First, a single ECG preprocessing and HRV analysis pipeline was applied across all recordings, with R-peak detection agreement assessed by a secondary detection path. Second, each task was interpreted using metrics appropriate to its physiological and signal-quality constraints: posture maps a steady-state autonomic set-point axis; breath-hold and walking reveal transient challenge–recovery trajectories; paced breathing modulates oscillation amplitude and spectral location. Third, the resulting metric interpretations are summarized in a task-specific validity table (Table **??**).

Table **??** maps each assignment question to the corresponding experiment, results section, and main-text figures.

TABLE I
ASSIGNMENT COVERAGE MAP (EEEC 1099 HW5)

Assignment Question	Experiment	Section	Main Figures
Q1: Postural baseline	E1 (A/B/C)	**??**	Fig. **??**–**??**
Q2: Transpulmonary effects	E2 (insp/exp)	**??**	Fig. **??**–**??**
Q3: Ambulatory recording	E3	**??**	Fig. **??**–**??**
Q4: Self-designed variation	E4: paced breathing	**??**	Fig. **??**–**??**
Cross-experiment synthesis	All	**??**	Fig. **??**, Table **??**

II. MATERIALS AND METHODS

A. Study Design and Protocol

One healthy adult male participant (age 22, no known cardiovascular or respiratory conditions) was instrumented with a VitalSigns VS-ECG single-lead chest-patch sensor sampling ECG at 500 Hz and triaxial accelerometry at 25 Hz. All 14 recordings were acquired in a single session on April 24, 2026, under quiet indoor conditions. Because every recording came from the same acquisition chain and was processed with a unified pipeline, between-condition differences are less likely to reflect changes in instrumentation.

The protocol comprised four experiment families (Table **??**):

1) **Postural baseline (E1).** After a 5-min rest, three consecutive 5-min recordings were obtained in supine (E1A), sitting (E1B), and standing (E1C) postures. The participant maintained relaxed spontaneous breathing throughout. Although the assignment specification requires only 30-s recordings per posture, this duration is too short for stable frequency-domain HRV estimation. With Welch's method and a Hann window, the effective spectral resolution is $\Delta f \approx 1.6/T$ [**?**]; for $T = 30$ s this yields $\Delta f \approx 0.053$ Hz, placing only $\sim$2 resolvable frequency bins within the entire LF band (0.04–0.15 Hz) and rendering LF power estimation unreliable. The Task Force guidelines explicitly recommend recordings of $\geq$5 min for short-term frequency-domain analysis [**?**]. We therefore extended each postural condition to 5 min (300 s), providing $\Delta f \approx 0.005$ Hz and $\sim$22 bins across the LF band, sufficient for stable Welch-based spectral estimation. The empirical effect of window duration on metric stability is quantified in Appendix A (Fig. **??**).
2) **Voluntary breath-hold (E2).** In a seated position, the participant performed two trials of maximum inspiratory breath-hold ($\sim$40 s each) followed by 40 s of recovery breathing, and two trials of maximum expiratory breath-hold with the same protocol. The assignment specifies 20-s breath-holds; we extended the target hold duration to $\sim$40 s to approximately double the number of cardiac cycles available during the apnoeic phase, to increase the number of cardiac cycles available for hold-phase time-domain metrics (HR slope, RMSSD) and to increase the spectral prominence of apnoea-related HRV changes within the per-trial PSD [**?**]. Inter-trial rest was provided until baseline HR was restored.
3) **Walking challenge (E3).** A 180-s continuous recording was segmented into 60 s seated baseline, 60 s walking at self-selected pace, and 60 s seated recovery.
4) **Paced breathing (E4).** *Self-designed variation (Assignment Q4).* This experiment served as the self-designed variation required by the assignment, formulated to test hypotheses **H4** and **H5** regarding the resonance question and LF/HF spectral-band migration under slow respiration. Five conditions were recorded in a seated position at fixed breathing rates of 12, 9, 6, 5, and 3 breaths/min, guided by a metronome. Each condition lasted approximately 4 min; the 3-breaths/min condition was extended to $\sim$5 min to ensure $\geq$14 complete breathing cycles. Inter-condition rest periods of $\geq$3 min were enforced to allow autonomic recovery and mitigate cumulative fatigue. *Limitation*: despite these rest periods, the five rates were presented in fixed descending order (12$\rightarrow$3), so progressive relaxation or training effects cannot be fully excluded without counterbalanced presentation.

TABLE II
EXPERIMENTAL PROTOCOL SUMMARY

Family	Condition	Duration (s)	Posture
E1 Postural	E1A Supine	300	Supine
	E1B Sitting	300	Sitting
	E1C Standing	300	Standing
E2 Breath-hold	Insp. $\times$2	120 each	Sitting
	Exp. $\times$2	120 each	Sitting
E3 Walking	Rest/Walk/Recovery	180	Sitting/Walk
E4 Paced	12, 9, 6, 5, 3/min	220–280	Sitting

B. Signal Processing

All processing was performed in Python 3.10 using a unified pipeline (`src/pipeline.py`). Raw ECG was corrected for baseline drift with a two-stage median filter (200 ms and 600 ms kernel widths), then filtered with a 50 Hz second-order IIR notch ($Q = 30$) to suppress power-line interference, followed by a 0.5–40 Hz fourth-order Butterworth bandpass (zero-phase via `sosfiltfilt`) [**?**].

R-peak detection followed two parallel paths to assess agreement between detection approaches (Fig. **??**):

- *Teaching path*: a Pan–Tompkins-style detector implemented in SciPy [**?**], providing a transparent, interpretable

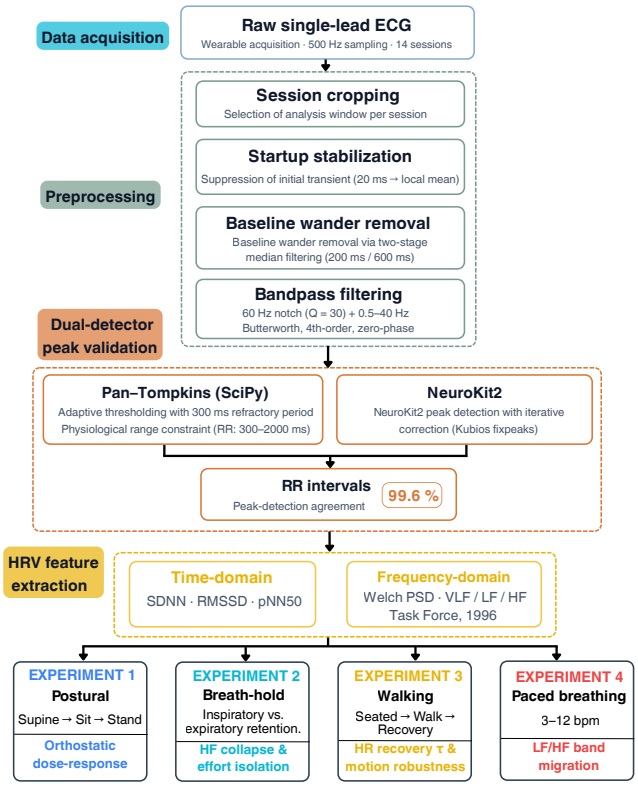

Fig. 1. End-to-end ECG analysis pipeline. Raw ECG is preprocessed with baseline correction and bandpass filtering, then processed through dual-path R-peak detection (SciPy teaching path and NeuroKit2 reference path) before experiment-specific HRV analysis.

reference.

- *Reference path*: NeuroKit2 `ecg_peaks` with Kubios-style artifact correction [**?**], serving as the primary path for reported results because it includes established artifact-correction routines for transient segments.

Pipeline validation on a 5-min steady-state recording (E1A) showed ≥99.5% peak-count agreement between paths, with mean RR difference <1.0 ms and RMSSD difference <2.7 ms.

C. HRV Analysis

HRV indices were computed following the Task Force standards [**?**]. RR intervals outside the 300–2000 ms physiological range were rejected. For steady-state conditions (E1, E4), the full analysis window (≥220 s) was used. Window-length sensitivity on the E1A supine recording (Appendix A, Fig. **??**) showed that sliding-window coefficient of variation fell below 10% for SDNN and below 5% for HF once segment duration reached ∼180 s, whereas 30–60 s windows exhibited much higher variability; this supports our use of full multi-minute windows—including 5-min postural baselines (E1) and the ≥220 s steady-state segments for paced breathing—for stable time- and frequency-domain HRV estimation.

Time domain. The following time-domain indices were computed from the edited NN interval series following Task Force definitions [**?**]:

- **SDNN**: the standard deviation of all NN intervals, reflecting total HRV encompassing both sympathetic and parasympathetic contributions. Because SDNN is algebraically dependent on recording length, values are comparable only across segments of equal duration [**?**].
- **RMSSD**: the root mean square of successive NN interval differences, selectively capturing short-term, beat-to-beat variability mediated predominantly by vagal (parasympathetic) efferent activity [**?**], [**?**]. Unlike frequency-domain HF power, RMSSD is insensitive to spectral-band boundary definitions and therefore remains less dependent on fixed spectral-band definitions when breathing frequency migrates across the LF/HF boundary.
- **pNN50**: the proportion of successive NN interval pairs differing by more than 50 ms, a complementary parasympathetic index highly correlated with RMSSD but employing a threshold criterion rather than a continuous measure [**?**].

Mean heart rate (HR) in beats per minute was computed as $60/\overline{\mathrm{NN}}$.

Frequency domain. The RR series was linearly detrended and resampled at 4 Hz via cubic interpolation [**?**]. Power spectral density (PSD) was estimated using Welch's method (256-point Hann window, 50% overlap). Band powers were integrated over the Task Force frequency ranges: VLF (0.003–0.04 Hz), LF (0.04–0.15 Hz), and HF (0.15–0.40 Hz). Normalized units were computed as $\mathrm{LF}_{\mathrm{n.u.}} = \mathrm{LF}/(\mathrm{TP} - \mathrm{VLF}) \times 100$.

Transient conditions (E2, E3). Because individual phases (hold, walk, recovery) lasted only 40–60 s, standard frequency-domain HRV was not computed per-phase [**?**]. Instead, we reported regime-based mean HR, RMSSD, HR range, and HR slope. For E2, frequency-domain HRV indices (HF power, LF/HF) were computed per trial using the same Welch method, and spectral comparison was performed against the E1B seated anchor.

Walking recovery kinetics. Post-walking HR recovery was fitted to a single-exponential decay model:

$$\mathrm{HR}(t) = \mathrm{HR}_\infty + (\mathrm{HR}_0 - \mathrm{HR}_\infty)\, e^{-t/\tau}, \tag{1}$$

using nonlinear least-squares (`scipy.optimize.curve_fit`) over the 60-s post-walking recovery window.

Motion contamination assessment. Walking-phase motion contamination was evaluated using two complementary metrics. First, Pearson correlation between absolute successive RR differences ($|\Delta\mathrm{RR}|$) and accelerometer vector magnitude was computed during the walking segment ($r = 0.26$). Second, magnitude-squared coherence (C_{xy}) between the walking ECG and accelerometer-derived motion magnitude was estimated to test whether a stable, phase-locked step-frequency component contaminated the cardiac signal. Because coherence captures phase-locked linear coupling better than amplitude correlation, both measures are interpreted jointly in Section **??**.

Paced-breathing frequency tracking. The dominant RR spectral peak was identified for each paced-breathing condition and compared against the imposed breathing frequency using ordinary least-squares regression. Slope, intercept, R^2, and 95% confidence intervals were reported; the null hypotheses slope $= 1$ and intercept $= 0$ were tested.

Respiratory-centered spectral analysis. To assess resonance independently of fixed LF/HF band boundaries, respiratory-centered RR spectral power was computed by integrating the Welch PSD within a narrow window around each imposed breathing frequency. The primary integration window was ± 0.015 Hz; sensitivity analyses were performed with ± 0.010 Hz and ± 0.020 Hz windows. This approach reduces reliance on fixed LF/HF bands when breathing frequency crosses the 0.15 Hz boundary, because the integration window tracks the respiratory peak rather than relying on fixed spectral bands. RR oscillation amplitude was additionally quantified as the distribution-based p95–p5 range, which is robust against cycle-level detection instability in the absence of a direct respiratory belt signal.

III. Results

*A. Postural Baseline Showed the Expected Orthostatic HRV Pattern (**H1 Supported**)*

Orthostatic loading was associated with monotonic changes in HRV metrics (Table **??**, Fig. **??**–**??**). From supine to standing, mean HR increased from 54.0 to 79.7 bpm, RMSSD decreased from 99.1 to 25.2 ms, HF power dropped from 3252.8 to 310.7 ms^2, and LF/HF rose from 0.28 to 1.44. These changes are consistent with progressive vagal withdrawal under orthostatic stress [**?**] and support **H1**.

The ECG power spectral density (Fig. **??**) further reveals the heartbeat fundamental frequency (f_0) and its harmonics ($2f_0$, $3f_0$), along with respiratory modulation visible as a low-frequency spectral peak, across all three postures.

Poincaré return maps (Fig. **??**) show the same reduction in beat-to-beat variability: the scatter cloud contracts progressively from supine to standing, consistent with the reduction in short-term vagal modulation measured by RMSSD.

The sitting baseline (E1B) was subsequently used as the posture-matched anchor for the seated breath-hold and walking experiments, reducing confounding by posture in between-experiment spectral comparisons.

For the ECG spectral analysis: (1) the main heartbeat frequency (f_0) and its harmonics ($2f_0$, $3f_0$) are clearly visible in the ECG PSD across all three postures (Fig. **??**); (2) the breathing frequency is identifiable as a low-frequency modulation peak; (3) successive heartbeat interval differences (Fig. **??**) show clearly larger amplitude in supine than in standing, visualizing the postural reduction in vagal modulation.

*B. Inspiratory and Expiratory Breath-Hold Produced Different HR Responses (**H2 Supported**)*

Inspiratory breath-hold produced larger HR responses than expiratory hold (Table **??**, Fig. **??**). Across two inspiratory trials, mean hold-phase HR rose to 66.9 and 73.5 bpm (ΔHR = +4.0 and +14.3 bpm relative to pre-hold baseline), whereas expiratory hold-phase HR remained near baseline (57.8 and 60.8 bpm; ΔHR = -0.9 and +1.9 bpm). The large inter-trial variability in the inspiratory condition (CV of pre-hold HR: 4.3% vs. 0.2% for expiratory) suggests that conscious effort and volitional strategy—in addition to respiratory cessation—shape the breath-hold response, supporting **H2**.

Spectral comparison against the E1B seated anchor showed a similar asymmetry (Fig. **??**). To reduce the influence of volitional effort, only the relaxed trial (Trial 2) was used for spectral quantification; the effort confound is addressed separately in Fig. **??**. Inspiratory hold HF power decreased markedly to 39.1 ms^2 (a 28.7-fold reduction from the anchor value of 1123 ms^2; -96.5%), whereas expiratory hold HF fell to 230.9 ms^2 (4.9-fold; -79.4%). Both conditions recovered to near-anchor levels within the 50 s post-hold window (inspiratory: 1207 ms^2, $\times 1.07$; expiratory: 1213 ms^2, $\times 1.08$), suggesting a transient perturbation with recovery toward the seated anchor.

TABLE III
POSTURAL BASELINE HRV SUMMARY

Metric	Supine	Sitting	Standing
Mean HR (bpm)	54.0	56.1	79.7
SDNN (ms)*	81.5	56.6	41.8
RMSSD (ms)	99.1	61.1	25.2
pNN50 (%)	62.2	43.5	2.8
Total Power (ms^2)	4321.9	2016.6	1047.8
LF (ms^2)	907.1	601.3	448.0
HF (ms^2)	3252.8	1122.6	310.7
LF n.u. (%)	21.8	34.9	59.0
HF n.u. (%)	78.2	65.1	41.0
LF/HF	0.28	0.54	1.44

*SDNN is duration-dependent [**?**]; all three conditions used equal 300-s windows.

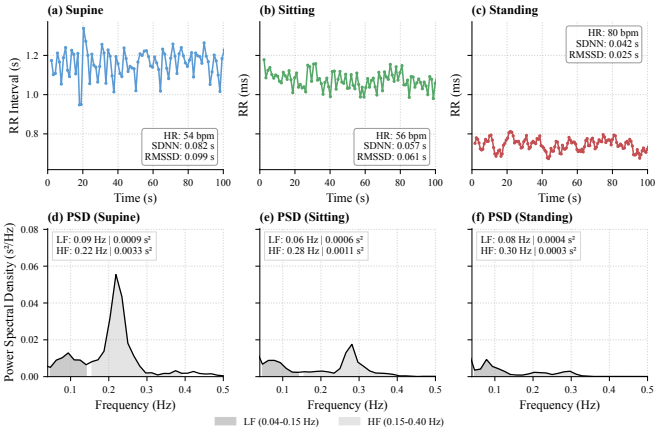

Fig. 2. Postural RR tachograms (top row) and RR interval power spectral density (bottom row) for supine, sitting, and standing conditions. RR intervals are displayed in seconds. Beat-to-beat successive differences are visible as the point-to-point variations in the tachogram, showing clearly larger amplitude in supine than in standing. Shaded regions indicate LF (orange) and HF (teal) bands per Task Force definitions [**?**]. HF power decreases monotonically from supine to standing, consistent with progressive vagal withdrawal.

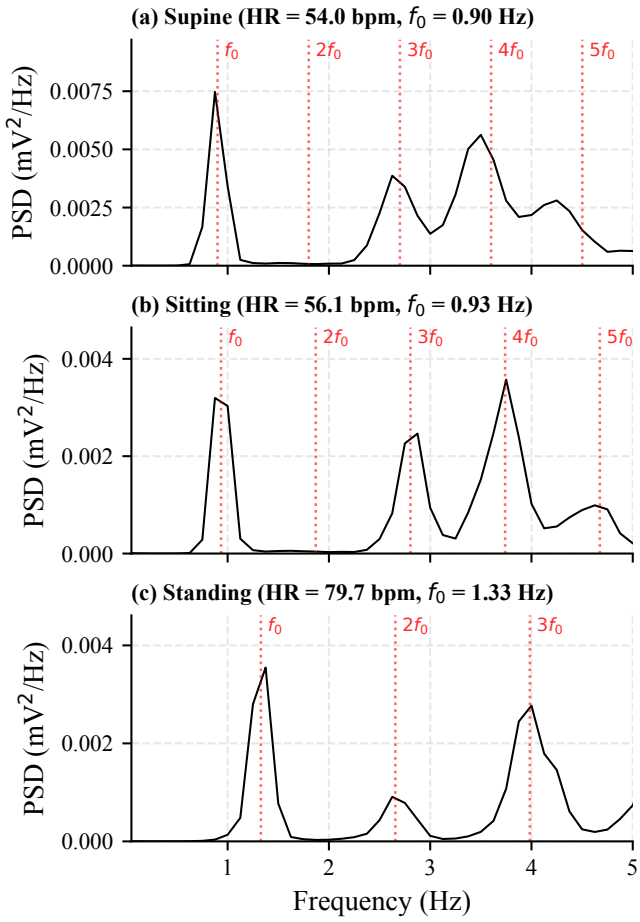

Fig. 3. ECG power spectral density (0.05–5 Hz) reveals heartbeat fundamental frequency, harmonics, and respiratory modulation across postures. The fundamental frequency f_0 shifts from ∼0.90 Hz (supine, HR = 54.0 bpm) to ∼1.33 Hz (standing, HR = 79.7 bpm). Harmonic peaks at $2f_0$ and $3f_0$ are clearly visible, consistent with the QRS complex morphology. The breathing frequency is identifiable as a low-frequency modulation peak in supine recordings. Direct response to assignment Q1: heartbeat frequency, harmonics, and breathing frequency are all observable.

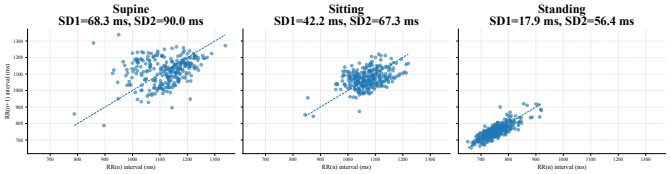

Fig. 4. Poincaré return maps (RR_{n+1} vs. RR_n) for supine, sitting, and standing conditions. The scatter cloud contracts progressively from supine to standing, reflecting reduced short-term vagal modulation consistent with the RMSSD decrease shown in Table **??**.

*C. Walking Increased HR but Limited Fine-Scale HRV Interpretation (**H3 Supported**)*

The walking challenge produced an acute cardiovascular response consistent with mild exercise (Table **??**, Figs. **??**, **??**). Mean HR rose from 60.8 bpm (seated) to 83.0 bpm (walking) and returned to 70.3 bpm during recovery, with a fitted exponential time constant $\tau = 6.8$ s (Eq. **??**). This rapid recovery is consistent with parasympathetic reactivation following low-intensity exercise [**?**].

However, walking-phase RMSSD (41.1 ms) should not be directly compared with resting RMSSD because walking introduces motion artifacts through multiple non-linear pathways—electrode-skin impedance fluctuations, trunk EMG, and cable sway—that cause random R-peak fiducial-point shifts and may inflate beat-to-beat variability indices. The raw walking ECG confirmed visible waveform degradation (Appendix D, Fig. **??**), and the walking-segment ECG spectrum showed elevated broadband energy in the 1–3 Hz locomotor band (Fig. **??**). The accelerometer-derived step frequency (1.46 Hz) fell within 6% of the walking heart-rate fundamental ($f_0 =$ 1.38 Hz at 83 bpm), creating a spectral overlap that limits reliable separation of cardiac and locomotor components by

TABLE IV
BREATH-HOLD TRIAL-LEVEL METRICS. TRIAL 1 = EFFORTFUL HOLD; TRIAL 2 = RELAXED HOLD (SEE FIG. **??**). HF POWER IS REPORTED FOR THE RELAXED TRIAL ONLY, BECAUSE THE OPPOSING HR-SLOPE DIRECTIONS BETWEEN TRIALS (−0.19 VS. +0.18 BPM/S FOR INSPIRATORY) VIOLATE THE STATIONARITY ASSUMPTION REQUIRED FOR ENSEMBLE SPECTRAL AVERAGING.

	Inspiratory		Expiratory	
Metric	Trial 1	Trial 2	Trial 1	Trial 2
Pre-hold HR (bpm)	62.9	59.2	58.8	58.9
Hold mean HR (bpm)	66.9	73.5	57.8	60.8
ΔHR (bpm)	+4.0	+14.3	−0.9	+1.9
HR slope (bpm/s)	−0.19	+0.18	−0.10	−0.11
Recovery HR (bpm)	79.2	64.8	59.2	57.4
HF power, relaxed trial only (vs. E1B anchor, 1123 ms^2):				
Hold HF (ms^2)	—	39.1	—	230.9
Recovery HF (ms^2)	—	1207	—	1213

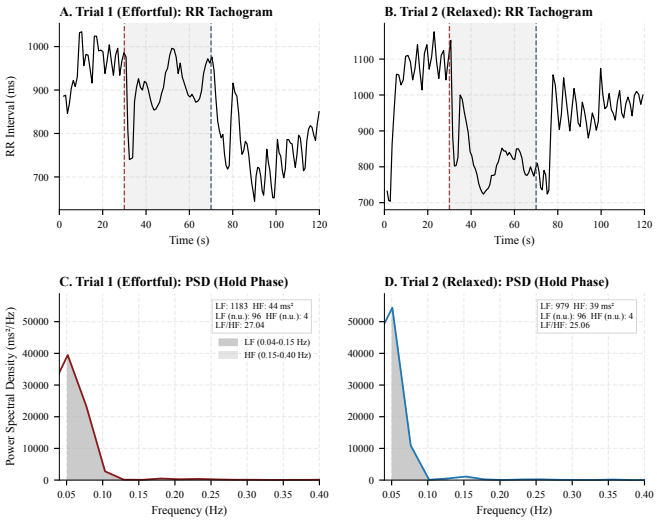

Fig. 5. Breath-hold effort comparison. Inspiratory hold (top panels) produces larger HR excursions and more variable response shapes than expiratory hold (bottom panels), suggesting that conscious effort may modulate the physiological response beyond the respiratory perturbation alone.

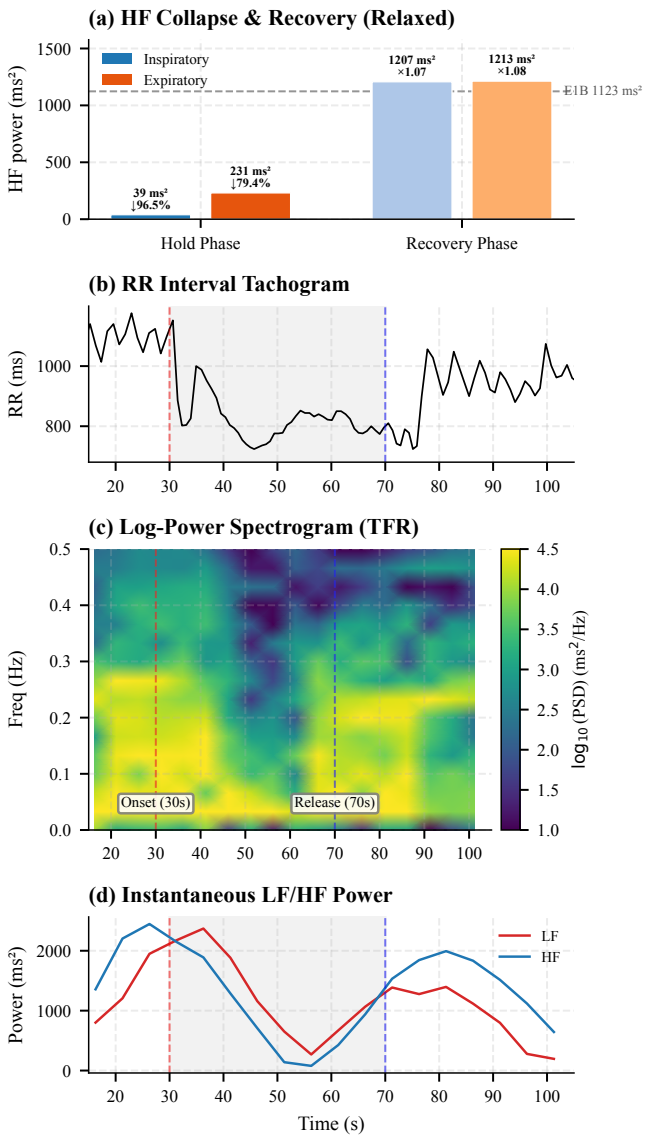

Fig. 6. Breath-hold spectral dynamics (relaxed trial only). (a) HF power collapses during hold relative to the E1B seated anchor (1123 ms^2): inspiratory hold shows a 28.7-fold reduction (-96.5%) versus a 4.9-fold reduction (-79.4%) for expiratory hold. Both conditions recover to near-anchor levels ($\times$1.07–1.08), indicating a transient vagal perturbation that returned toward baseline. (b)–(d) Representative inspiratory-relaxed trial showing the RR tachogram, time–frequency spectrogram, and instantaneous LF/HF power trajectories.

linear filtering alone.

We computed magnitude-squared coherence to test for phase-locked, linear coupling between locomotor mechanics and the cardiac signal. Coherence was low both at the step peak ($C_{xy} = 0.04$) and across the 1–3 Hz cadence band (mean $C_{xy} = 0.06$, max 0.20), providing little evidence for a stable phase-locked linear contamination pathway; Pearson correlation between $|\Delta\mathrm{RR}|$ and accelerometer magnitude was likewise weak ($r = 0.26$). Low coherence does not, however, exclude non-linear artifacts such as broadband baseline wander

TABLE V
WALKING CHALLENGE SEGMENT METRICS

Segment	HR (bpm)	RMSSD (ms)	HR slope	Note
Seated	60.8	54.4	-0.06	
Walking	83.0	41.1	-0.12	Motion†
Recovery	70.3	46.9	-0.13	$\tau = 6.8$ s
E1B anchor	56.1	61.1	—	Seated ref.

†Walking-phase HRV is motion-contaminated; interpret with caution.

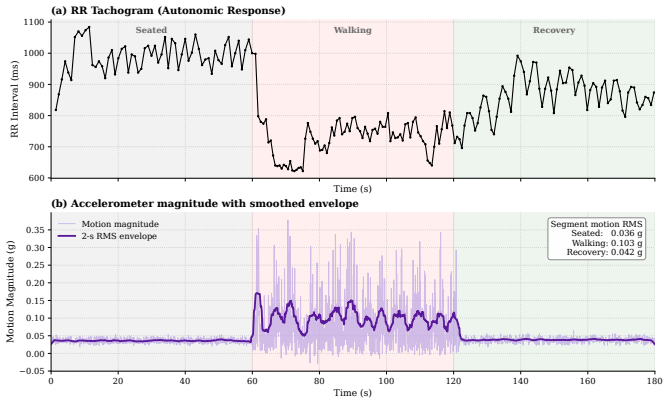

Fig. 7. Walking experiment overview. (a) RR tachogram showing the autonomic response across seated, walking, and recovery segments. HR rises sharply at walk onset and decays exponentially during recovery ($\tau = 6.8$ s). (b) Accelerometer magnitude with 2-s RMS envelope. Segment-level motion RMS confirmed minimal motion during seated (0.036 g) and recovery (0.042 g), with elevated but moderate motion during walking (0.103 g). Time-domain Pearson correlation between $|\Delta\mathrm{RR}|$ and accelerometer magnitude was $r = 0.26$.

or stochastic fiducial-point jitter. Combined with the cadence-proximate spectral energy in 1–3 Hz (Fig. **??**); walking-phase fine-scale HRV was therefore treated as motion-limited.

Mean HR and recovery kinetics—which depend on slow trends robust to fiducial jitter—remain physiologically interpretable, while fine-scale HRV indices such as RMSSD are motion-limited. The walking-phase HRV was accordingly flagged as **unreliable (motion)** in the summary table, consistent with **H3**.

For completeness: (1) motion interference is observable in the walking-phase RR series and ECG PSD, manifesting as elevated spectral energy in the 1–3 Hz band coincident with the walking cadence identified independently from the accelerometer (Fig. **??**, Fig. **??**); (2) the post-walking rest segment showed progressive lengthening of RR intervals with $\tau = 6.8$ s, consistent with parasympathetic reactivation following low-intensity exercise.

*D. Paced Breathing: Entrainment Manipulation Check, LF/HF Band Migration (**H5**), and Resonance (**H4**)*

Experiment 4A provided the most structured manipulation in this study and yielded three observations.

1) Frequency Entrainment Was Quantitatively Verified: The measured RR spectral peak closely tracked the imposed breathing frequency (Fig. **??**). Linear regression yielded

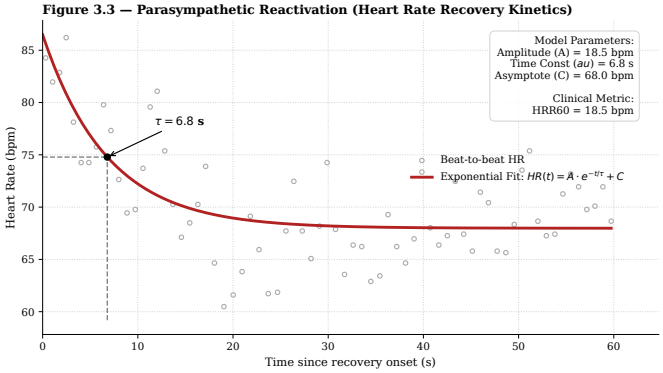

Fig. 8. Post-walking heart rate recovery fitted to a single-exponential decay (Eq. **??**). The time constant $\tau = 6.8$ s and $\mathrm{HRR}_{60} = 18.5$ bpm indicate rapid parasympathetic reactivation consistent with low-intensity exercise. Grey circles: beat-to-beat HR; red curve: fitted model $\mathrm{HR}(t) = A \cdot e^{-t/\tau} + C$.

slope $= 1.022$ (95% CI: 0.947–1.097), intercept $= -0.002$ Hz (95% CI: -0.012 to 0.008), $R^2 = 0.998$, and residual SD $= 0.003$ Hz. Neither the slope-=1 nor the intercept-=0 null hypothesis was rejected, indicating that the manipulation was effective and that the paced-breathing protocol produced respiratory entrainment of the cardiac cycle (manipulation check).

2) The Primary Effect Was Amplitude Expansion, Not Heart-Rate Reduction: Across the five paced conditions, mean HR varied only narrowly (57.0–60.1 bpm), whereas SDNN rose from 48.7 to 131.4 ms, total power increased from 1754 to 10,493 ms^2, and peak-to-trough RR amplitude grew from 157 to 406 ms (Table **??**, Fig. **??**). In this dataset, slow breathing primarily expanded oscillatory amplitude rather than shifting the mean chronotropic operating point.

*3) Respiratory-Centered Spectral Power Did Not Peak at 6 Breaths/Min (**H4** Supported):* Classical resonance biofeedback theory predicts a peak in RR oscillation amplitude near 6 breaths/min ($\sim$0.1 Hz) [**?**]. To test this without relying on fixed LF/HF bands or cycle-level peak-to-trough detection—which can be unstable during very slow breathing without a direct respiratory belt signal—we computed respiratory-centered RR spectral power by integrating the Welch PSD within $\pm$0.015 Hz of each imposed breathing frequency. Both respiratory-centered power and distribution-based RR amplitude (p95–p5) increased monotonically from 12 breaths/min to 3 breaths/min (1003 to 13,460 ms^2 and 157 to 406 ms, respectively; Fig. **??**), without forming an inverted-U shape centered at 6 breaths/min. Sensitivity analysis using $\pm$0.010 Hz and $\pm$0.020 Hz integration windows showed that the same monotonic pattern was observed across all three bandwidths (Appendix E, Fig. **??**). A classical 6-breaths/min resonance maximum was *not observed* in this participant under the present fixed-order protocol (**H4** supported).

*4) LF/HF Inflation Dissociated from RMSSD, Supporting Spectral-Band Migration (**H5** Supported):* LF/HF rose dramatically from 0.22 (12/min) to 11.54 (6/min) and plateaued near 17.5 at the slowest rates, yet RMSSD remained in a narrow range of 48–66 ms throughout (Fig. **??**). This dissociation is consistent with **H5**: when the respiratory peak crosses the 0.15 Hz LF/HF boundary, the resulting LF/HF increase reflects respiratory spectral-band migration—not a sympathovagal shift [**?**]. RMSSD, being computed entirely in the time domain, is unaffected by frequency-band assignment and therefore provides a more stable vagal index under slow paced breathing. Together with the respiratory-centered analysis above, these results indicate that slow paced breathing inflates LF/HF because the dominant respiratory peak shifted from HF into LF, rather than by shifting autonomic balance.

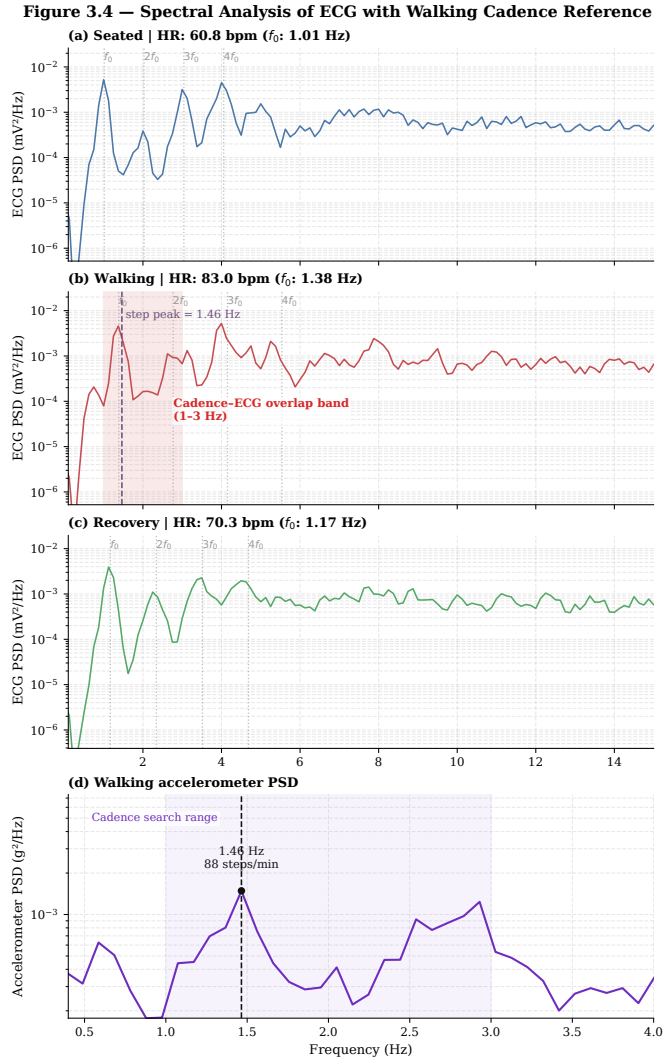

Fig. 9. Walking-phase ECG power spectral density showing elevated energy in the 1–3 Hz locomotor band coincident with the walking cadence. Motion artifacts contaminate the spectral content during ambulation, limiting fine-scale HRV interpretation.

E. Cross-Experiment Comparison Required Task-Specific Metrics

When all experiments are viewed together (Fig. **??**), two distinct patterns were observed. *Posture* primarily shifted the autonomic operating point along a trajectory of increasing

TABLE VI
PACED BREATHING: FREQUENCY TRACKING AND HRV METRICS

	12/min	9/min	6/min	5/min	3/min
Expected (Hz)	0.200	0.150	0.100	0.083	0.050
Measured (Hz)	0.203	0.148	0.102	0.086	0.047
$\lvert\Delta f\rvert$ (Hz)	0.003	0.002	0.002	0.003	0.003
HR (bpm)	59.4	57.0	60.2	59.9	58.1
SDNN (ms)*	48.7	78.0	91.3	95.9	131.4
RMSSD (ms)	47.7	64.9	60.4	53.9	55.3
TP (ms^2)	1754	2381	6087	8240	10,493
LF (ms^2)	279	953	5470	7634	8980
HF (ms^2)	1242	1129	474	437	514
LF/HF	0.22	0.84	11.54	17.45	17.47
RR amp. (ms)	157	240	293	297	406

*SDNN is duration-dependent [?]; condition durations ranged from 220 to 280 s.

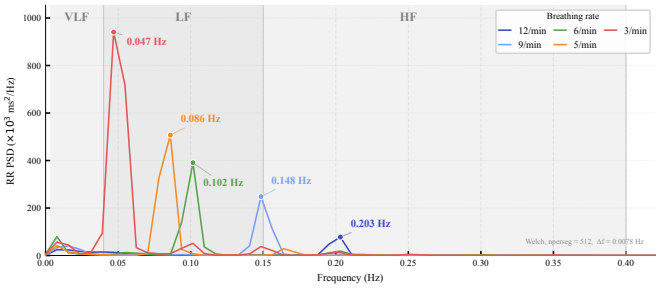

Fig. 11. RR interval PSD for five paced-breathing conditions. The dominant spectral peak migrates leftward with decreasing breathing rate, crossing the LF/HF boundary (dashed vertical line at 0.15 Hz) between 12/min and 6/min. Peak amplitude increases monotonically with slower breathing.

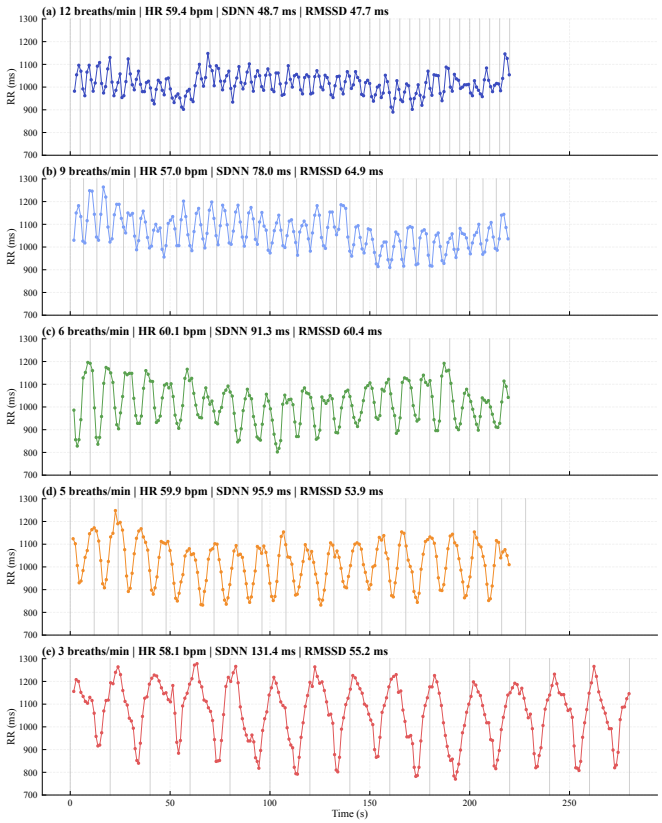

Fig. 10. Paced-breathing RR tachograms for five breathing rates (12, 9, 6, 5, 3 breaths/min). Light grey vertical lines indicate metronome-cued inhalation onsets used by the participant to pace breathing. RR oscillation amplitude increases progressively with slower breathing, while mean RR interval (and thus HR) remains relatively stable across conditions.

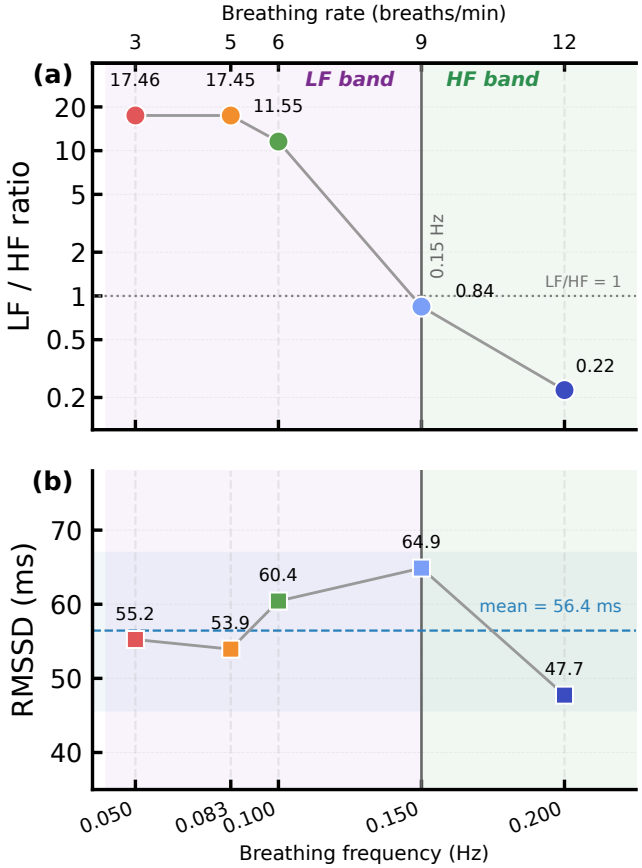

Fig. 12. LF/HF ratio (left axis) vs. RMSSD (right axis) across paced-breathing conditions. LF/HF inflates dramatically below 9 breaths/min while RMSSD remains stable, indicating that the LF/HF increase reflects spectral-band migration rather than sympathovagal reversal.

HR and decreasing RMSSD (supine → standing), with total power declining monotonically. *Paced breathing*, by contrast, maintained nearly stable mean HR while expanding total power by a factor of 6.0 (1754→10,493 ms^2) and SDNN by a factor of 2.7.

These represent different types of physiological modulation and cannot be ranked on a single LF/HF axis. The breath-hold and walking paradigms, meanwhile, are best described as challenge–recovery trajectories: inspiratory hold showed the stronger HR rise, expiratory recovery showed the largest RMSSD rebound, and walking recovery followed an exponential time course.

These results favor reporting cross-experiment summaries using mean HR, RMSSD, total power, and explicit validity annotations (e.g., "motion-limited," "band-shifted") rather than

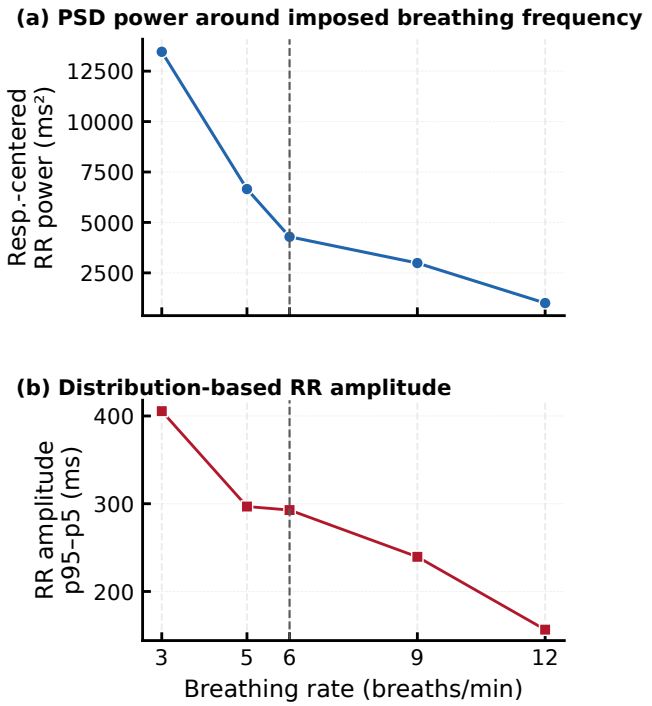

Fig. 13. Resonance assessment via respiratory-centered metrics. (a) RR PSD power integrated within ±0.015 Hz of each imposed breathing frequency. (b) Distribution-based RR amplitude (p95–p5). Both measures increase monotonically from 12 to 3 breaths/min; neither exhibits an inverted-U maximum at 6 breaths/min (dashed vertical line). A classical 6-breaths/min resonance peak was not observed in this participant under the present fixed-order protocol.

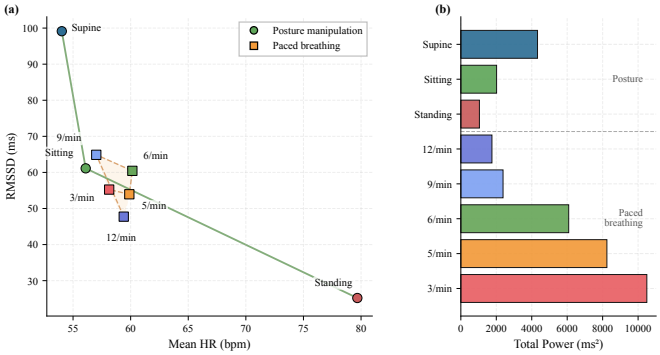

Fig. 14. Cross-experiment autonomic map. Steady-state conditions are plotted by LF and HF power (log scale) with mean HR on a twin axis. Posture shifts the operating point along the HR axis; paced breathing expands total spectral power at nearly constant HR. Transient experiments (breath-hold, walking) appear in separate annotations.

reducing all conditions to a single frequency-domain scalar.

IV. Discussion

A. Context Determines Interpretation

The main observation is that HRV indices did not retain the same physiological interpretation across task contexts. The same LF/HF ratio that separated supine from standing (0.28 vs. 1.44) became physiologically uninterpretable under slow paced breathing (17.47), where it was largely explained by the mechanical relocation of the respiratory peak from HF into LF. This result aligns with longstanding critiques that LF/HF "does not accurately measure cardiac sympatho-vagal balance" [**?**], [**?**] and provides a within-participant example: the same participant generated a 60-fold range in LF/HF (0.28–17.47) through postural and respiratory manipulations alone, without any pharmacological or pathological intervention.

B. Paced Breathing: Amplitude, Not Rate

In this dataset, the most prominent effect of slow breathing was amplitude expansion of the respiratory sinus arrhythmia, not reduction of mean heart rate. In our data, mean HR varied by only 3.1 bpm across the five paced conditions (57.0–60.1 bpm), whereas peak-to-trough RR amplitude increased by 159% (157–406 ms). This finding is consistent with the view that slow breathing amplifies the gain of the cardiorespiratory coupling loop—the baroreflex, respiratory gating of vagal efferents [**?**], and mechanical intrathoracic pressure oscillations—without a large change in mean HR [**?**], [**?**].

The absence of a 6-breaths/min resonance maximum in this participant warrants cautious interpretation. Classical resonance theory [**?**] predicts that RR oscillation amplitude peaks when breathing frequency matches the cardiovascular system's natural resonance, typically near 0.1 Hz. Using respiratory-centered RR spectral power—a metric that tracks the imposed breathing frequency rather than relying on fixed LF/HF bands—we found that both spectral power and RR amplitude (p95–p5) increased monotonically down to 3 breaths/min (0.05 Hz). The monotonic pattern was unchanged across three integration bandwidths (±0.010, ±0.015, ±0.020 Hz). However, this is a single-participant observation under a fixed descending presentation order (12→3 breaths/min), so the monotonic increase may partly reflect progressive relaxation, deeper tidal volumes at slower rates, or cumulative training effects rather than an absence of resonance. A randomized or counterbalanced multi-participant design with concurrent respiratory monitoring would be needed to distinguish these possibilities from a true resonance-frequency shift.

C. Practical Implications for HRV Reporting

Based on this analysis, three reporting practices are recommended for similar multi-condition HRV protocols:

1) Report LF/HF only when breathing rate is confirmed to lie within the HF band (>0.15 Hz, i.e., >9 breaths/min). Otherwise, use RMSSD or total power as the primary summary.
2) Annotate each condition with a validity note indicating whether the standard interpretation applies (e.g., "standard," "band-shifted," "motion-limited").
3) Use posture-matched anchors when comparing transient tasks against baselines, to avoid confounding postural and task-specific autonomic effects.

Table **??** lists the metric interpretation used for each condition in this study.

TABLE VII
TASK-SPECIFIC HRV METRIC VALIDITY SUMMARY

Condition	HRV valid?	Primary metric	Caveat
Supine/sitting/ standing (E1)	Yes	RMSSD, HF, LF/HF	Steady-state
Breath-hold (E2)	Partial	HR response, recovery, HF	Transient; short-segment PSD
Walking (E3)	Motion-limited	Mean HR, recovery τ	RMSSD unreliable
Paced $\geq$9/min (E4)	Mostly valid	RMSSD, HF, LF/HF	Resp. peak in or near HF band
Paced $\leq$6/min (E4)	Band-shifted	RMSSD, TP, RR ampl., resp.-centered power	LF/HF invalid as sympathovagal index; use resp.-centered power for resonance

V. LIMITATIONS

This lab study has five limitations. First, it is a single-participant within-subject investigation; the findings illustrate metric-behavior principles under controlled within-subject manipulation but cannot claim population generalizability. We deliberately framed this as a methodological lab demonstration rather than an inferential study.

Second, the paced-breathing conditions were presented in fixed descending order (12→3 breaths/min). Inter-condition rest periods of $\geq$3 min were enforced to allow autonomic recovery, but progressive relaxation or training effects cannot be fully excluded without counterbalanced or randomized presentation order.

Third, no direct respiratory signal (respiratory belt or impedance pneumography) was recorded. Respiratory entrainment was inferred from the RR spectral peak; ECG-derived respiration (EDR) validation is recommended for future iterations of this lab.

Fourth, walking-phase HRV is motion-contaminated, and fine-scale beat-to-beat metrics during ambulation should be interpreted with explicit motion-limited validity caveats. Magnitude-squared coherence between the walking ECG and accelerometer was low ($C_{xy} = 0.04$ at the step peak), consistent with absence of strong phase-locked linear coupling but not excluding broadband or non-linear artifact; the combination of high accelerometer motion and cadence-proximate spectral energy warrants caution for fine-scale HRV indices.

Taken as a methodological demonstration, the within-subject design shows how HRV metrics changed across four task contexts.

VI. CONCLUSION

Using a unified single-lead ECG pipeline across four paradigms, this lab study showed distinct HRV response patterns across posture, breath-hold, walking, and paced breathing. Posture shifts the autonomic operating point (**H1**). Breath-hold reveals asymmetric inspiratory–expiratory dynamics (**H2**). Walking increased cardiovascular load but limits fine-scale HRV interpretability (**H3**). Within the tested range of paced breathing, no classical 6-breaths/min resonance maximum was observed (**H4**). Slow paced breathing showed that LF/HF can reflect band migration rather than a sympathovagal index (**H5**). Overall, the results support a task-specific interpretation of HRV metrics. Under slow paced breathing, mean HR, RMSSD, and total power provided the most interpretable cross-condition summary in this dataset.

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APPENDIX A
DATA QUALITY AND PIPELINE VALIDATION

A. Dual-Path R-Peak Detection Agreement

Dual-path R-peak detection (SciPy teaching path vs. NeuroKit2 reference path) was validated on the E1A 5-min supine recording. Peak count agreement was 99.6% (269 vs. 268 peaks). The mean RR difference was 0.96 ms (0.09%), SDNN difference was 1.65 ms (2.1%), and RMSSD difference was 2.69 ms (2.7%). Across all 8 steady-state sessions (E1A–C, E4 12–3 breaths/min), peak-count agreement exceeded 99.5%.

Bland–Altman analysis on mean NN intervals (Fig. **??**b) yielded a bias of -0.43 ms with 95% limits of agreement of $[-1.86, +1.00]$ ms, well below the temporal resolution required for HRV interpretation. RMSSD showed a slightly larger bias (Fig. **??**c, $\Delta = 1.24$ ms; 95% limits $[-2.21, +4.68]$ ms), reflecting RMSSD's higher sensitivity to single-beat detection differences. Per-session artifact rates (Fig. **??**a) remained below 2.5% in all recordings, with the highest values concentrated in the breath-hold (E2) sessions where transient HR shifts mildly increase the proportion of corrected beats; steady-state sessions remained below 1%.

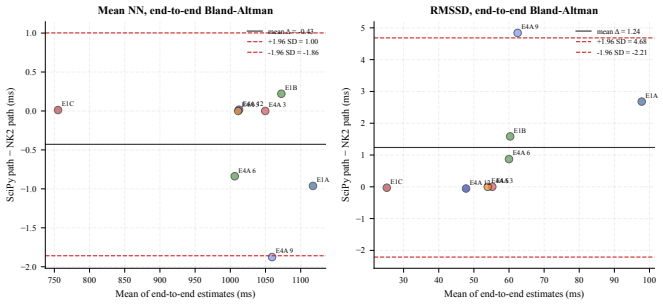

Fig. 15. Dual-path validation summary. (a) Per-session artifact rate. (b) Bland–Altman comparison of mean NN intervals between the SciPy teaching path and the NeuroKit2 reference path. (c) Bland–Altman comparison of RMSSD between the two paths.

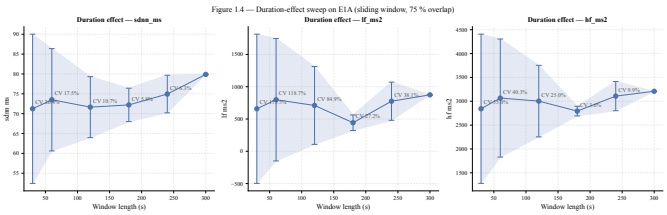

Fig. 16. Effect of analysis-window duration on HRV metric stability in the E1A supine recording. SDNN and HF stabilize substantially by ∼180–250 s, whereas LF remains comparatively variable.

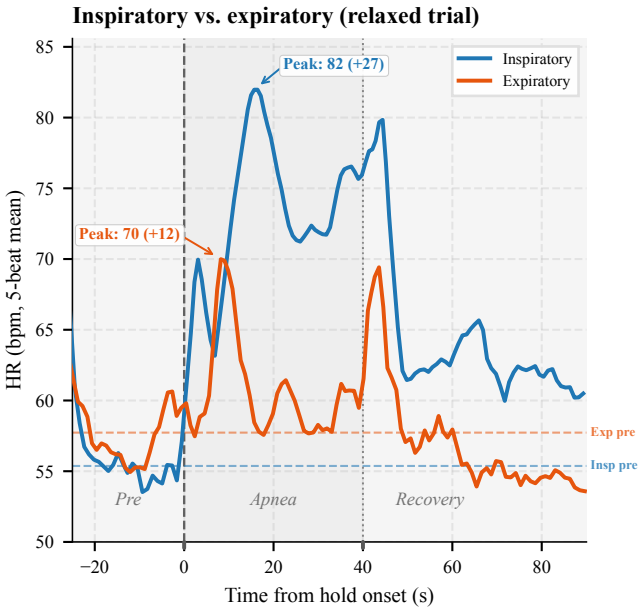

Fig. 17. Inspiratory vs. expiratory relaxed-trial breath-hold overlay, time-aligned to hold onset. The inspiratory hold produces the earlier and larger HR excursion, isolating the physiological effect of lung inflation apart from voluntary effort.

B. Analysis-Window Duration Sensitivity

Figure **??** reports the coefficient of variation (CV) of SDNN, LF, and HF as a function of sliding-window duration on the E1A supine recording (75% overlap). SDNN stabilized rapidly: CV fell from 26.4% at 50 s to 5.8% at 250 s. HF showed lower variability, dropping to 3.6% CV by 250 s. LF remained the noisiest band (CV $>$ 27% even at 250 s), reflecting its sensitivity to slow trends and sub-band drift. These results support the use of full multi-minute windows—5 min for postural baselines (E1) and $\geq$220 s for steady-state paced breathing (E4)—rather than the shorter 30–60 s windows commonly used for transient analyses. This empirical observation is consistent with the theoretical spectral-resolution constraint discussed in Section **??**: a 30-s Hann window yields $\Delta f \approx 0.053$ Hz, placing only ∼2 bins inside the LF band and precluding reliable LF power estimation [**?**].

C. Recording Inventory and Artifact Burden

All 14 recordings used in this study passed quality control. The peak-correction burden distribution (Fig. **??**a) shows a clear separation between steady-state recordings (artifact rate $<$1%), transient breath-hold trials (∼1–2.5%), and the walking segment (∼1.5%, expected given motion artifacts). No recording required exclusion.

Appendix C
Breath-Hold Trial-Level Detail

The main-text Fig. **??** shows the two inspiratory trials to highlight the effort/relaxation contrast used to evaluate H2. This appendix provides the complementary inspiratory-vs.-expiratory overlay restricted to the relaxed trial, thereby reducing the voluntary-strain confound present in Trial 1 and focusing on the respiratory-cessation effect.

Figure **??** overlays the relaxed inspiratory trial (Trial 2) against the relaxed expiratory trial, time-aligned to hold onset. The inspiratory trace reaches a peak HR of 82 bpm (+27 above pre-hold), whereas the expiratory trace reaches only 70 bpm (+12). The earlier and larger HR rise in the inspiratory condition may be consistent with elevated intrathoracic pressure compressing venous return during sustained lung inflation, complementing the spectral evidence shown in main-text Fig. **??**.

The figure also clarifies the temporal asymmetry of recovery. After release, both traces relax toward baseline over the subsequent recovery window, but the inspiratory hold shows a steeper immediate descent from its higher peak. This pattern is consistent with the observation from Table **??** that the dominant difference between the two relaxed holds is not hold duration, but the magnitude of the circulatory perturbation induced by lung volume at the onset and during sustained apnea.

Appendix D
Walking Detail Figures

This appendix provides supplementary views of the walking experiment (E3) data. The main-text Figs. **??**–**??** cover the autonomic response, recovery kinetics, and cadence-coincident ECG spectral contamination. The figures below add signal-quality context needed to interpret why mean HR and recov-

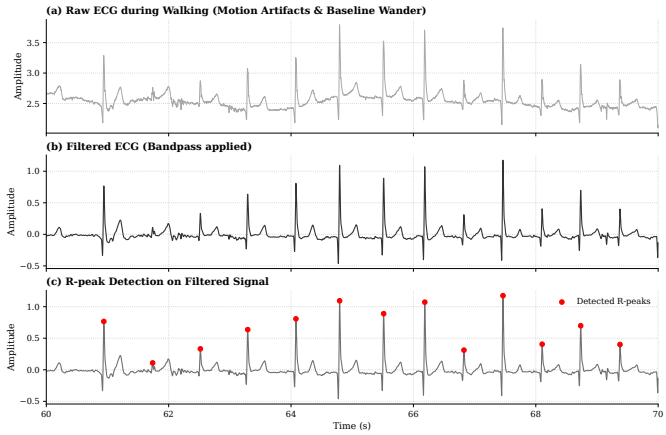

Fig. 18. ECG signal processing during the walking segment (60–70 s). (a) Raw ECG showing motion artifacts and baseline wander characteristic of ambulatory recording. (b) Bandpass-filtered signal with substantially reduced low-frequency drift. (c) R-peak detection on the filtered signal. Despite visible motion contamination, the NeuroKit2 detector successfully identified all QRS complexes in this segment.

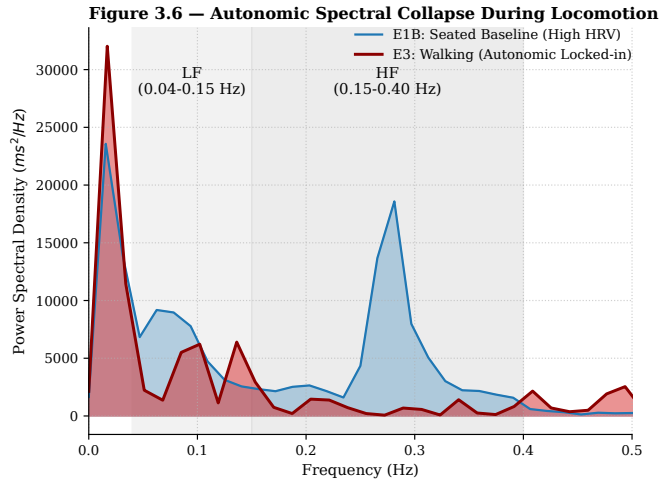

Fig. 19. Walking vs. seated-baseline RR power spectral density contrast. Walking shows near-collapse of HF power together with broadening of the LF region, consistent with sympathetic activation plus motion-related spectral spreading.

ery remain trustworthy while walking-phase fine-scale HRV should be treated cautiously.

Figure **??** shows a representative 10-s window of walking-phase ECG before and after preprocessing, demonstrating that despite visible motion artifacts and baseline wander in the raw signal, the bandpass-filtered output still supports reliable R-peak detection. This suggests that the motion-limited HRV interpretation in Section **??** reflects measurement uncertainty during ambulation, rather than obvious detector failure.

Figure **??** compares the seated-baseline (E1B) and walking RR power spectra on a common axis. The walking spectrum shows strongly attenuated HF power and a broadened LF region, consistent with reduced beat-to-beat variability under elevated sympathetic drive combined with motion-induced spectral spreading. In that sense, the figure complements Fig. **??**: the main-text panel shows where locomotor contamination enters the ECG spectrum, while the appendix contrast shows the resulting flattening of physiologically interpretable RR structure.

These two panels suggest a bounded conclusion for E3: walking still permits extraction of average rate and recovery timing, but it degrades the assumptions required for high-confidence interpretation of short-lag variability metrics. This is why the walking-phase RMSSD value is reported for completeness but not used as a cross-condition autonomic comparator.

Appendix E
Paced Breathing Resonance: Sensitivity Analysis

The respiratory-centered RR spectral power metric introduced in Section **??** depends on the choice of integration bandwidth around each imposed breathing frequency. To verify that our central conclusion—absence of a 6-breaths/min resonance maximum—is not an artifact of bandwidth selection, we repeated the analysis with three integration windows: ± 0.010, ± 0.015 (primary), and ± 0.020 Hz.

Figure **??** shows that all three bandwidths yield the same monotonic decrease in respiratory-centered power from 3 to 12 breaths/min, with no isolated peak at 6 breaths/min. The narrower ± 0.010 Hz window shows slightly lower absolute power because some respiratory-band energy falls outside the integration limits at higher breathing rates where peaks are broader, but the qualitative shape—and notably, the absence of an inverted-U—is preserved.

This sensitivity analysis supports the interpretation that the negative result is unlikely to be explained solely by bandwidth choice, although fixed-order effects remain possible. The paced-breathing manipulation produced frequency tracking and large amplitude changes without revealing a classical resonance maximum at 6 breaths/min.

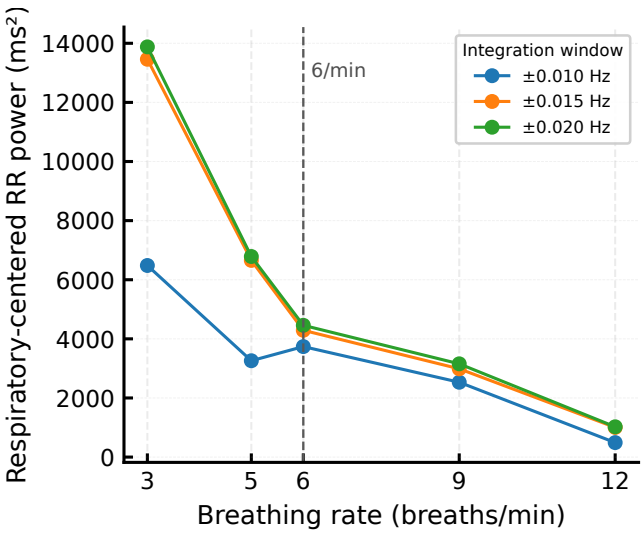

Fig. 20. Sensitivity analysis of respiratory-centered RR spectral power across three integration windows (± 0.010, ± 0.015, ± 0.020 Hz) around each imposed breathing frequency. All three bandwidths show monotonic increase from 12 to 3 breaths/min with no isolated peak at 6 breaths/min (dashed vertical line).